DIFFERENT TECHNIQUES FOR GLUCOSE ESTIMATION AND THEIR RELEVANCE WITH DIABETES MELLITUS

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Learning objectives

Students should be able to

- Determine different lab techniques for glucose estimation
- Corelate glucose estimation methods with clinical condition specially diabetes mellitus.

Diabetes Mellitus

- Diabetes is a condition that happens when blood sugar (glucose) is too high.
- It develops when pancreas doesn't make enough insulin or any at all, or when the body isn't responding to the effects of insulin properly.
- Diabetes affects people of all ages.
- Glucose (sugar) mainly comes from carbohydrates food and drinks. It's body's go-to source of energy. Blood carries glucose to all body's cells to use for energy.
- When glucose is in the bloodstream, it needs help — a "key" — to reach its final destination. This key is insulin . pancreas isn't making enough insulin or the body isn't using it properly, glucose builds up in your bloodstream, causing high blood sugar (hyperglycemia)

- Type 2 Diabetes: Insulin resistance
- Type 1 Diabetes: autoimmune destruction of cells in pancreas that produces insulin.
- Gestation diabetes : Develops during pregnancy

Symptoms of Diabetes



Increased thirst.



Slow-healing cuts and sores.



Fatigue.



Blurred vision.



Frequent urination.



Unexplained weight loss.

Cleveland Clinic

Diabetes-related ketoacidosis (DKA):

- This complication mainly affects people with Type 1 diabetes or undiagnosed T1D.
- It happens when body doesn't have enough insulin. If the body doesn't have insulin, it can't use glucose for energy, so it breaks down fat instead.
- This process eventually releases substances called ketones, which turn your blood acidic.
- This causes labored breathing, vomiting and loss of consciousness. DKA requires immediate medical treatment.

Symptoms of diabetes include:

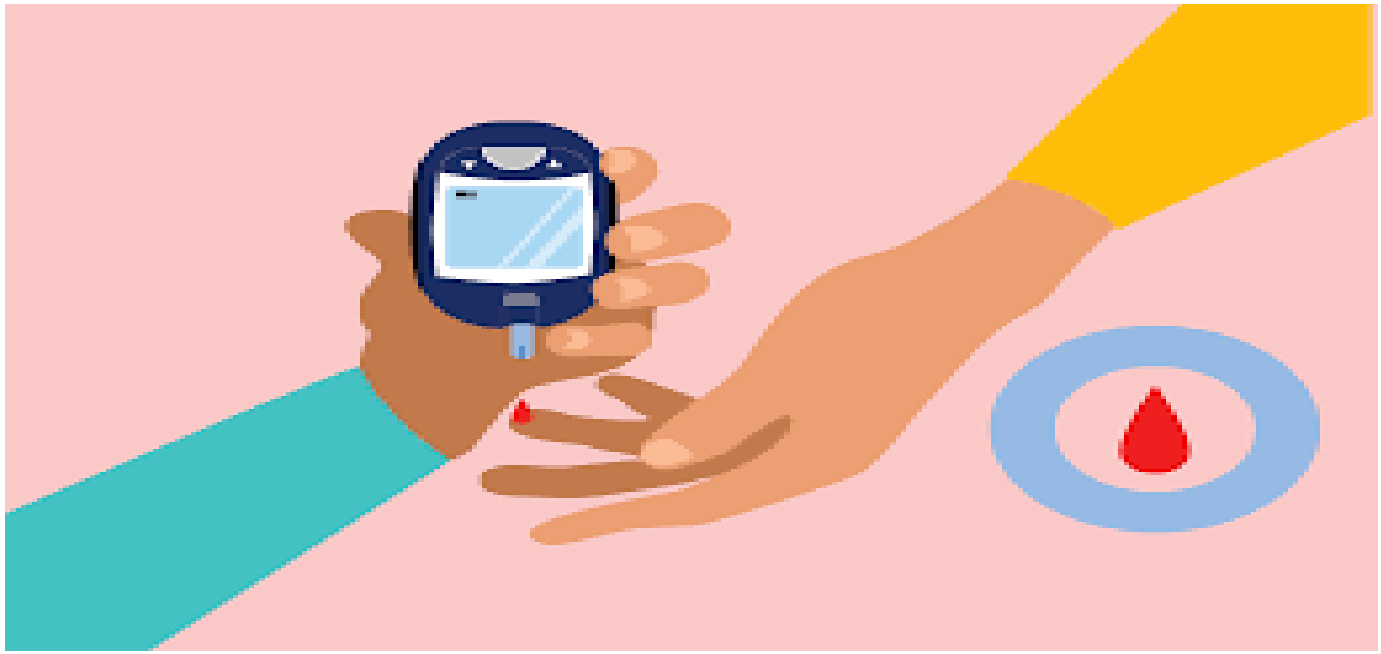
- Increased thirst (polydipsia) and dry mouth.
- Frequent urination
- Fatigue.
- Blurred vision
- Unexplained weight loss
- Numbness or tingling in your hands or feet.
- Slow-healing sores or cuts.
- Frequent skin infections

Fasting Blood Glucose Test: For this test, patienst doesnot eat or drink anything except water (fast) for at least eight hours before the test. As food can greatly affect blood sugar, this test allows to see baseline blood sugar.

Random blood Test : “Random” means that one can get this test at any time, regardless of if they have fasted.

HbA1C : This test, also called HbA1C or glycated hemoglobin test, provides your average blood glucose level over the past two to three months.

To screen for and diagnose gestational diabetes, providers order an oral glucose tolerance test



Type of test	In-range (mg/dL)	Prediabetes (mg/dL)	Diabetes (mg/L)
Fasting blood glucose test	Less than 100.	100 to 125.	126 or higher.
Random blood glucose test	N/A.	N/A.	200 or higher (with classic symptoms of hyperglycemia or hyperglycemic crisis).
A1c	Less than 5.7%.	5.7% to 6.4%.	6.5% or higher.

Glucose Tolerance Test

- A glucose tolerance test (GTT) is a test that checks to see if the body can use and store glucose (sugar) as it should.
- There are several variations of the glucose tolerance test. The most common is the oral glucose tolerance test (OGTT).
- A one-hour Glucose challenge test during pregnancy is a routine test to screen for gestational diabetes. a three-hour OGTT is performed to officially diagnose or rule out the condition.

a glucose tolerance test to help screen for or diagnose any of the following conditions:

- Diabetes (usually prediabetes , gestational diabetes)
- Insulin resistance
- Reactive hypoglycemia (low blood sugar).
- Acromegaly
- Rare carbohydrate metabolism disorders

Method of Glucose Tolerance Test

- The specific steps of a glucose tolerance test can vary depending on the type.
- As an example of the steps, the most common type: the two-hour oral glucose tolerance test (OGTT).
- In most cases, we start the test in the morning — usually between 7:00 and 9:00 a.m.

- The test typically involves the following steps:
- A provider will take a blood sample from vein. This is to measure your fasting blood glucose.
- A provider will give the candidate a glucose liquid to drink. It’s usually 75 grams of glucose dissolved in 250 to 300 mL of water. The provider will instruct to drink it all within five minutes. The dose is different for children and pregnant women The glucose drink tastes like flat soda, which may taste unpleasant for some people.
- A provider will take a blood sample from the candiate’s vein two hours after drinking the glucose to check blood glucose. Depending on the reason for the test, they may draw blood several times every 30 to 60 minutes after the candidate has drunk the solution.

One Hour Glucose Tolerance Test =

- This is typically a glucose challenge test to screen for gestational diabetes during pregnancy.
- The candidate drinks a liquid with 50 grams of glucose and then a provider draws your blood to check your blood sugar one hour later.
- You typically don't need to fast for this test

Three Hour Glucose Tolerance Test =

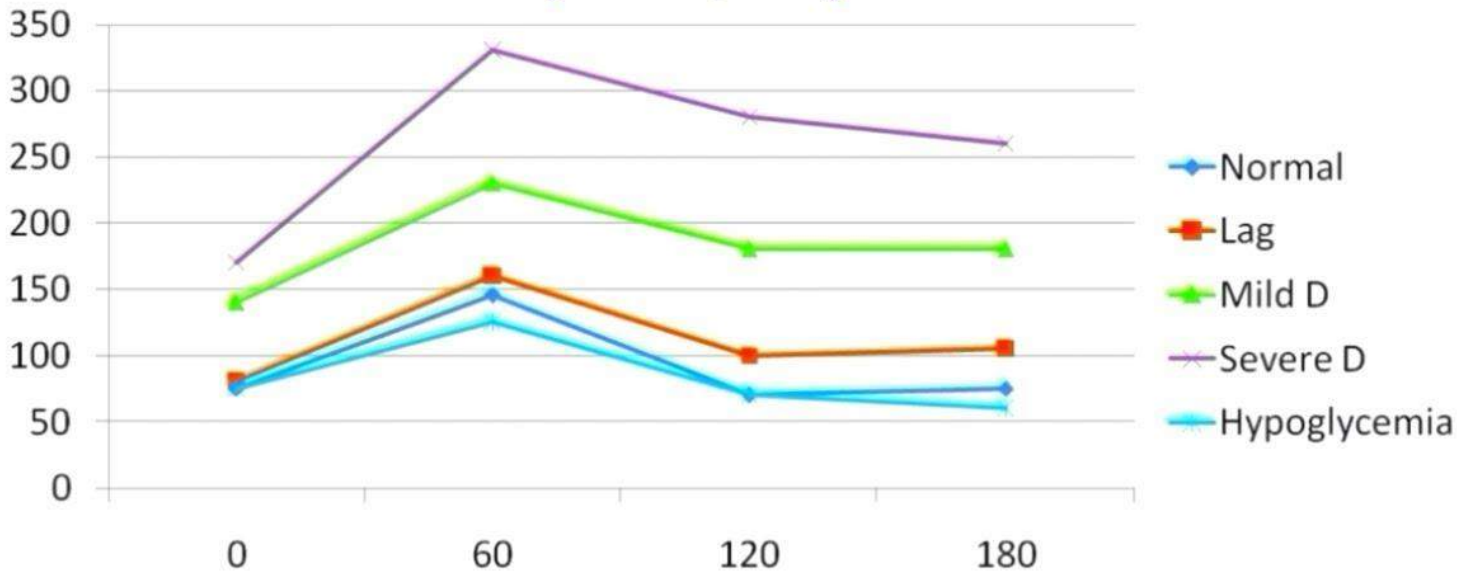
- You'll have blood draws before you drink the glucose liquid (fasting), and at one, two and three hours after drinking it.

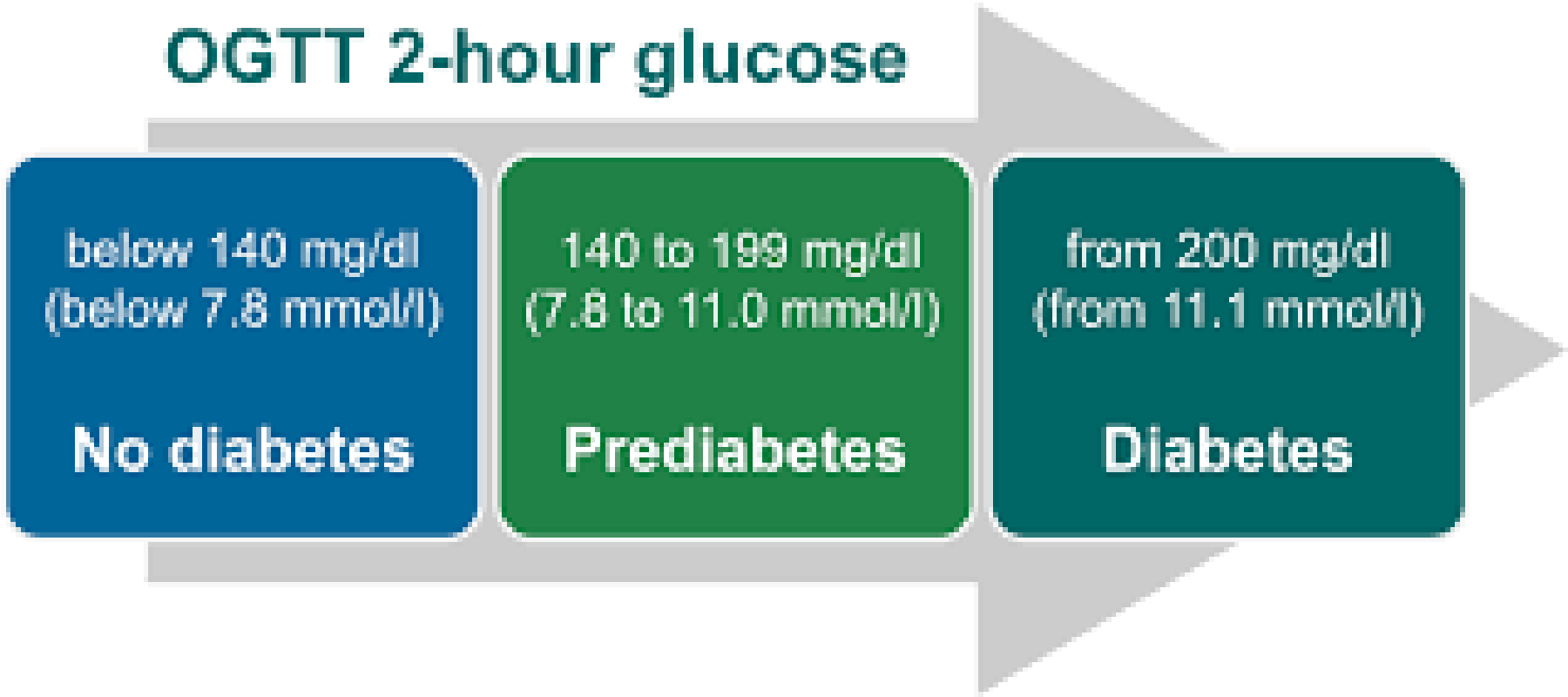
Glucose Tolerance Test for Acromegaly =

- During this test, your provider will draw blood samples to check your human chorionic gonadotropin (HCG) level (not your blood sugar).
- After you drink the glucose liquid, they'll likely draw your blood every 30 minutes for two hours at 30-, 60-, 90- and 120-minute intervals.

Oral Glucose Tolerance Test

OGTT, GTT




OGTT 2-hour glucose
below 140 mg/dl
(below 7.8 mmol/l)
No diabetes
140 to 199 mg/dl
(7.8 to 11.0 mmol/l)
Prediabetes
from 200 mg/dl
(from 11.1 mmol/l)
Diabetes

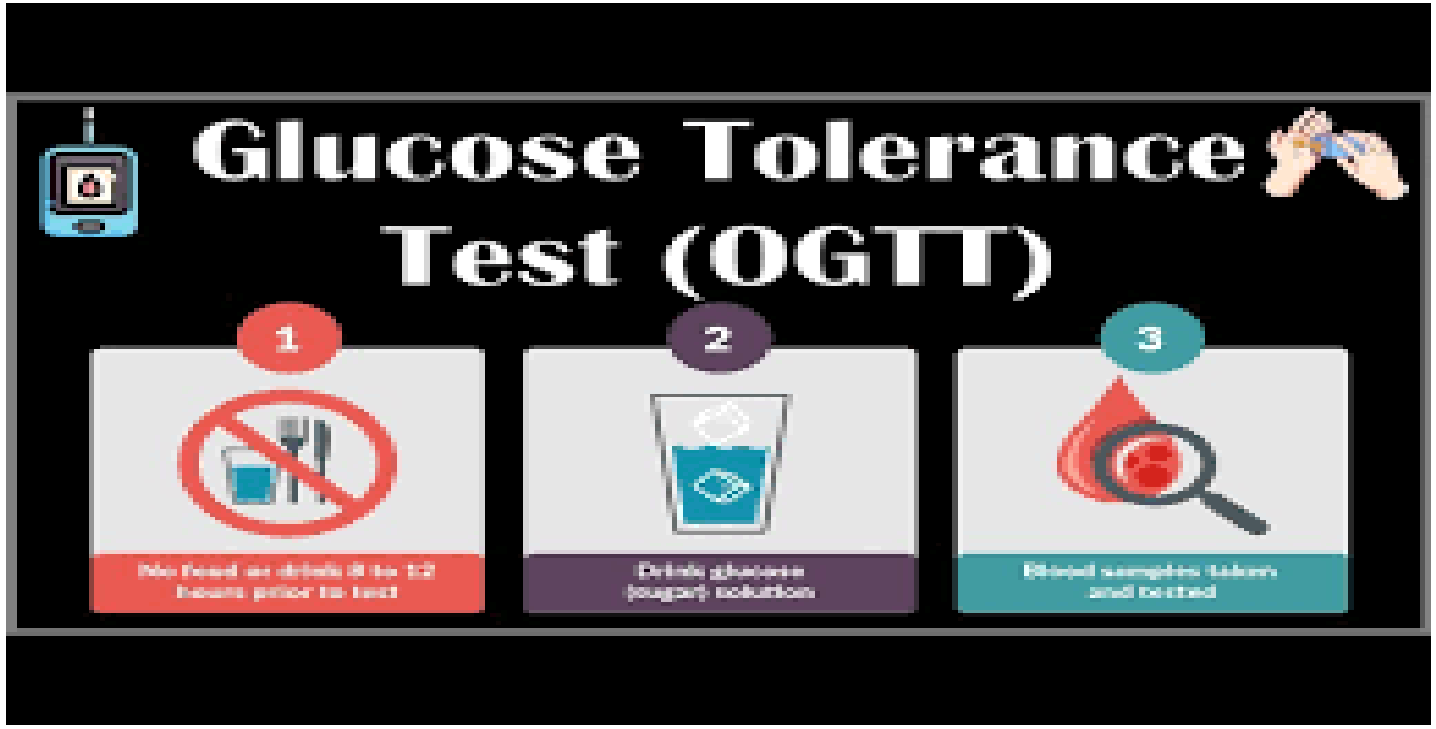
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Glucose Tolerance Test

Normal persons

Fasting	60-100 mg/dl
After 1 hr	200 mg/dl
After 2 hrs	140 mg/dl

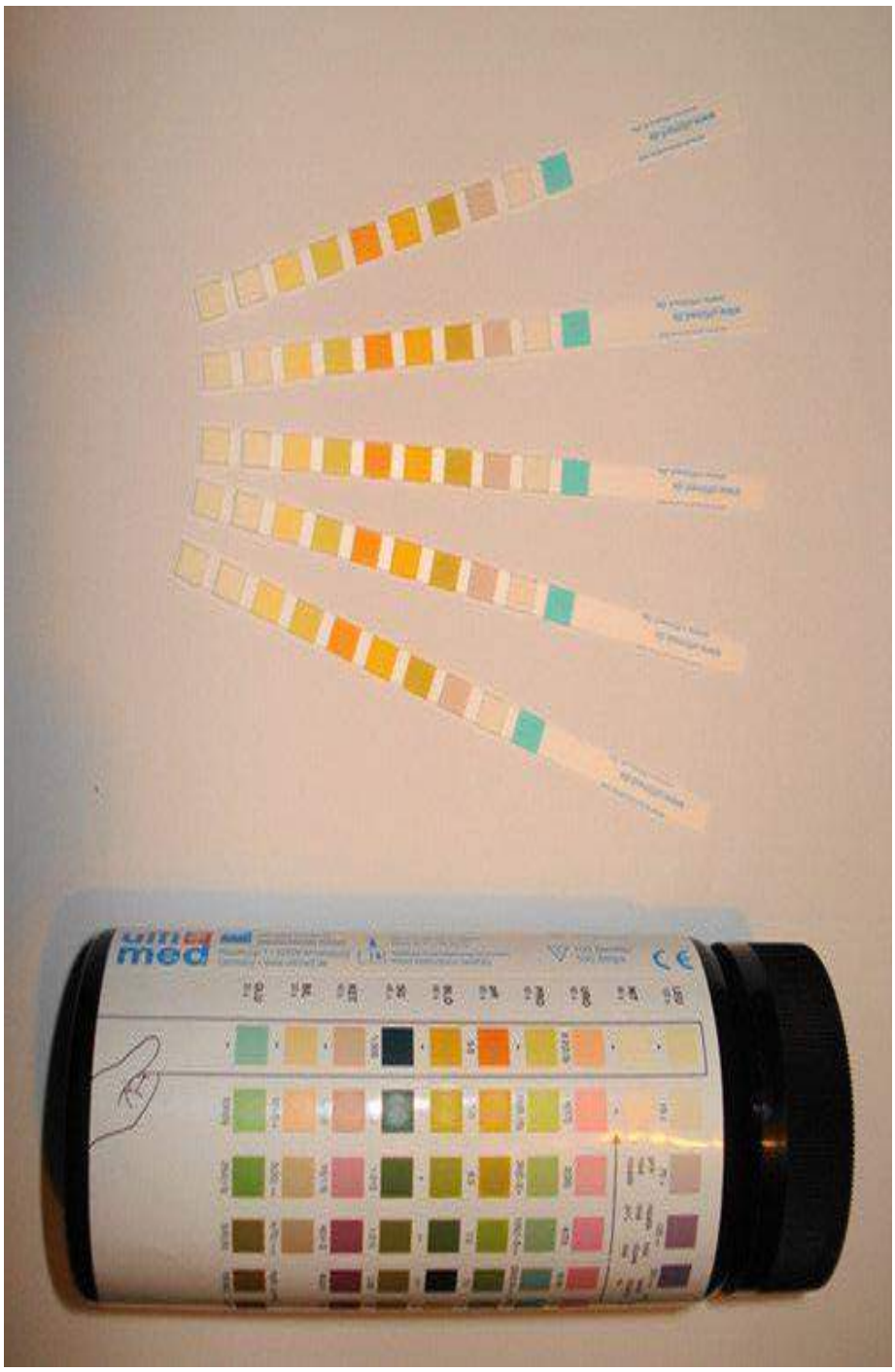

Glucose Tolerance Test (OGTT)
1
2
3
Drinks glucose (sugar) solution

Urine Dipstick

- A urine dipstick test tests the urine using a special strip of paper that is dipped into a sample of urine.
- It is the quickest way to test urine. It involves dipping a specially treated paper strip into a sample of urine.
- The results are usually available within 60-120 seconds.
- The sample doesn't need to be sent to a laboratory for a urine dipstick test although if the test is abnormal, the sample may be sent to a laboratory for further tests.



Leukocytes	Increased concentration may indicate infection in kidneys and urinary tracts. In bacterial urinary tract infection leukocytes are in most cases found in the urine.
Nitrite	Positive test result may indicate urinary tract infection. Gram-negative bacteria converts nitrate to nitrite in the bladder.
Urobilinogen	Increased concentration may indicate liver cell damage or increased bilirubin excretion to intestines.
Protein	May indicate kidney disease.
PH	Indicates urine acidity level. Acidic urine may be caused by kidney disease, but also by diet.
Blood	Increased concentration may indicate infection or disease in kidney and bladder.
Specific gravity	Increased concentration may give information on the kidneys' ability to concentrate urine in relation to plasma.
Ketones	Increased concentration may be associated with diabetes, low-carb diets or starvation.
Bilirubin	Increased concentration may indicate liver damage.
Glucose	Increased concentration may indicate diabetes mellitus.


urine
med

Method of urine Dipstick

- Taking a urine sample as above.
- Dipping one of the sticks into the sample, making sure all the test zones on the stick are submerged.
- Removing the stick, tapping to remove excess urine and holding it horizontally so urine doesn't drip.
- After the relevant time (usually between 30 seconds and 2 minutes), holding the dipstick up against the colour chart on the dipstick bottle.
- Each test will be colour-coded and there will be several different shades of that colour. A darker shade indicates that there is more of the relevant substance - blood, glucose etc - in the urine.
- Recording the result.

Purpose of urine Dipstick

- A urine dipstick test is done for lots of different reasons, such as:
- A routine check during pregnancy to check for glucose, protein or signs of bacteria.
- To screen for diabetes
- If a urine infection is suspected.
- To look for blood in urine (hematuria).
- It may be done in people with some types of abdominal or back pain to look for kidney or bladder problems.

Enzymatic Method

- **1. Glucose Oxidase-Peroxidase (GOD-POD) Method**

Principle:
Glucose is oxidized by the enzyme glucose oxidase to gluconic acid and hydrogen peroxide. The hydrogen peroxide then reacts with a chromogen in the presence of peroxidase to form a colored compound, which can be measured spectrophotometrically.

Reactions:

- $\text{Glucose} + O_2 \xrightarrow{\text{(Glucose Oxidase)}} \text{Gluconic acid} + H_2O_2$
- $H_2O_2 + \text{Chromogen} \xrightarrow{\text{(Peroxidase)}} \text{Colored product} + H_2O$

- **Common chromogens:** o-Dianisidine, ABTS, 4-aminoantipyrine with phenol
  Measurement: Absorbance at 505–550 nm (depends on chromogen used)

- **Applications:** Widely used in clinical laboratories and glucose meters.

Hexokinase Method

- **Principle:**
  Glucose is phosphorylated by hexokinase in the presence of ATP to form glucose-6-phosphate (G6P), which is then oxidized by glucose-6-phosphate dehydrogenase (G6PD) with the reduction of NAD^+ or $NADP^+$ to NADH/NADPH. The increase in absorbance due to NADH/NADPH formation is measured.

- **Reactions:**
- Glucose + ATP —(Hexokinase)→ Glucose-6-phosphate + ADP
- G6P + $NAD(P)^+$ —(G6PD)→ 6-Phosphogluconate + NAD(P)H + H^+

Measurement: Absorbance at 340 nm (due to NADH/NADPH)

- **Advantages:** High specificity and accuracy
  Applications: Reference method in laboratories

Glucose Dehydrogenase (GDH) Method

- There are several variants depending on the cofactor used (NAD^+, $NADP^+$, FAD, PQQ):

a. GDH-NAD^+ Method

Reaction

- Glucose + NAD^+ —(GDH)→ Gluconolactone + NADH

- **Measurement:** Absorbance at 340 nm
- **b. GDH-PQQ Method**
- Uses pyrroloquinoline quinone as a coenzyme; more common in glucose biosensors.
- **Advantages:**

- Rapid and suitable for test strips in point-of-care devices
- Not oxygen-dependent (unlike GOD method)
- Works in low oxygen environments

Comparison Table

Method	Enzymes Used	Detection Method	Specificity	Use Case
GOD-POD	Glucose oxidase, peroxidase	Colorimetric	High	Routine glucose testing
Hexokinase	Hexokinase, G6PD	Spectrophotometric (340 nm)	Very High	Reference method in labs
GDH (NAD/PQQ)	Glucose dehydrogenase	Spectrophotometric or electrochemical	High	Test strips, POC glucose meters

- Thank you